

SOS/VOD and early endothelial complications

Week 10 · EBMT Handbook Ch. 49 and Ch. 42

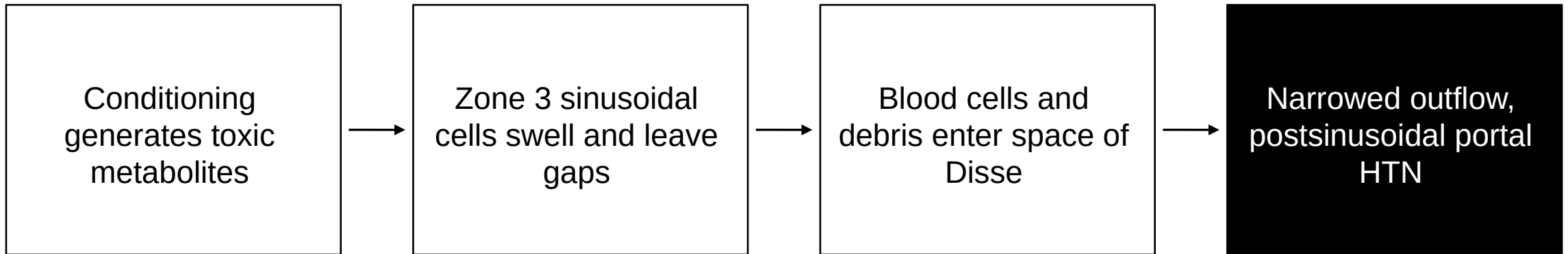
Hsieh-Ting Lin, MD

Department of Medical Oncology, Koo Foundation Sun Yat-Sen Cancer Center

Four parts, from the liver to the whole endothelium

- | | | |
|----------|---|----------|
| 1 | SOS/VOD in adults
Mechanism, risk factors, EBMT 2023 criteria, severity grading, defibrotide | 10.5 min |
| 2 | SOS/VOD in children
Pediatric EBMT criteria, refractory thrombocytopenia, prophylaxis debate | 2 min |
| 3 | FOS, CLS, engraftment and pre-engraftment syndrome
Timing, triggers, diagnostic criteria, treatment | 3.5 min |
| 4 | TA-TMA, PRES and three board questions
Harmonized criteria, risk strata, complement blockade | 10 min |

SOS/VOD begins with injury to zone 3 sinusoidal endothelial cells



Adults: incidence about 5–15%. Most cases resolve within weeks, but severe SOS/VOD carries a mortality above 80%. Day-0 EASIX is a promising marker of high risk.

Five factors carry the highest SOS/VOD risk; many others are modifiable

Highest relative risk	Other unmodifiable	Modifiable
Older patient age	Second HCT; disease beyond CR2	Myeloablative conditioning
Female receiving norethisterone	Karnofsky <90%; metabolic syndrome	Oral or high-dose busulfan
Prior gemtuzumab or inotuzumab ozogamicin	Iron overload (>1,000 ng/mL)	High-dose treosulfan or TBI
Serum bilirubin >26 µmol/L	Transaminases >2.5× ULN; hepatotoxic drugs	Unrelated or HLA-mismatched donor
Fibrosis, cirrhosis, active viral hepatitis	Abdominal or hepatic irradiation	Sirolimus or MTX with a CNI; parenteral nutrition

Children add: age <2 years, HLH, osteopetrosis, JMML, neuroblastoma, prior SOS/VOD.

EBMT 2023 adds a probable category so that treatment is not delayed

Finding	Probable	Clinical	Proven
Bilirubin ≥ 2 mg/dL	Counts	Mandatory	—
Painful hepatomegaly	Counts	Counts	—
Weight gain $> 5\%$	Counts	Counts	—
Ascites	Counts	Counts	—
Ultrasound or elastography suggestive	Counts	—	—
Histology, or HVPg ≥ 10 mmHg	—	—	Sufficient
Rule	Any 2 of 5	Bilirubin + 2 of 3	Either one

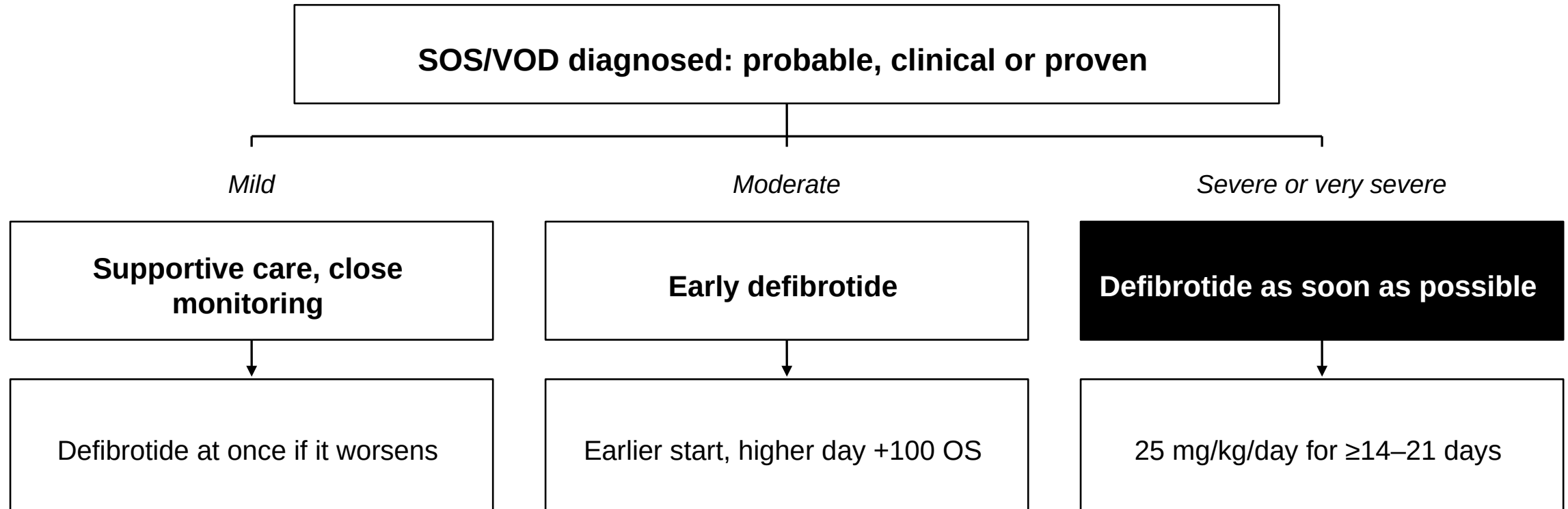
Onset $\leq$ day 21 = classical; $>$ day 21 = late-onset. Same criteria; exclude other causes.

Grade by the worst category with ≥ 2 criteria; ≥ 2 risk factors upgrade

Parameter	Mild	Moderate	Severe	Very severe (MOD)
Time since symptoms	>7 days	5–7 days	≤ 4 days	Any time
Bilirubin, mg/dL	≥ 2 and < 3	≥ 3 and < 5	≥ 5 and < 8	≥ 8
Bilirubin kinetics	—	—	Doubling in 48 h	—
Transaminases	$\leq 2\times$ normal	$> 2\text{--}5\times$ normal	$> 5\text{--}8\times$ normal	$> 8\times$ normal
Weight increase	—	—	$\geq 5\%$	$\geq 10\%$
Creatinine vs baseline	Baseline	$< 1.5\times$	≥ 1.5 and $< 2\times$	$\geq 2\times$ or MOD

Validation (n=203, Yoon 2019): day +100 TRM 36.7% if very severe vs 2.7–8.3% in other grades.

Defibrotide is started early from moderate SOS/VOD upward



UDCA is the recommended prophylaxis; defibrotide prophylaxis is disputed

Measure	Data in the chapter	Position
Avoid modifiable risk factors	Conditioning, donor, GVHD prophylaxis, nutrition	Crucial
Ursodeoxycholic acid	From conditioning to day +90 (Ruutu 2014)	Recommended (adults)
Defibrotide, pediatric RCT (2012)	360 high-risk children: 20% vs 12%, p=0.049	Effective
Defibrotide, HARMONY (Grupp 2021)	372, half adults: SOS-free survival 67% vs 73%, p=0.85	Not effective
Children, high risk	pEBMT criteria already allow earlier defibrotide	Open question

Children need pEBMT criteria: no bilirubin threshold and no time limit

pEBMT criterion (any 2 of 5)	Why it differs from adults
Weight gain on 3 days despite diuretics, or >5% in 72 h	Replaces fixed 3% or 5% cut-offs
Rising bilirubin on 3 days, or ≥ 2 mg/dL within 72 h	About 30% of pediatric SOS/VOD is anicteric
Hepatomegaly above baseline, best on imaging	Baseline ultrasound before HCT
Ascites above baseline, best on imaging	Often preexistent in HLH, osteopetrosis, TDT
Unexplained transfusion-refractory thrombocytopenia*	Earliest and most sensitive sign

*** ≥ 1 weight-adjusted platelet transfusion/day. Szmit 2020: diagnosis 3 days earlier than Seattle.**

FOS, CLS, ES and pES overlap; timing and trigger separate them

Syndrome	When and who	Defining features	Management
FOS	Around day 0; up to 60%	Weight gain, dyspnoea, high CVP; no cardiac or renal failure	Diuretics; no impact on TRM
CLS	Mainly children; 5.4%	Weight >3%/24 h, edema, no response to furosemide ≥1 mg/kg	Stop G-CSF; steroids; hemodynamic support
ES	Peri-engraftment; auto 7–59%, allo 10–25%	Noninfectious fever, rash, hypoxia; CRP >6 mg/dL	Stop G-CSF, antibiotics; then methyl-PDN 1 mg/kg q12h × 3 d
pES	Cord blood (MAC); about day +7; 20–70%	Like ES, 10–11 days before engraftment; more fluid retention	Same as ES

EASIX = LDH × creatinine / platelets. At day 0 it predicts SOS/VOD and early fluid retention.

Use Maiolino criteria after auto-HCT and Spitzer criteria after allo-HCT

Criterion	Spitzer 2001	Maiolino 2003	Grant 2020
Required	3 M, or 2 M + ≥ 1 m	1 M + 1 m	2 M, or 1 M + ≥ 2 m
Window around engraftment	96 h before to 96 h after	24 h before, any time after	96 h before to 24 h after
Noninfectious fever	M	M	M
Rash >25% of body surface	M	m	M
Pulmonary edema or hypoxia	M	m	m
Weight gain >2.5% (Maiolino >3%)	m	m	m
Hepatic/renal dysfunction; encephalopathy	m	—	—
Diarrhea, ≥ 2 episodes	—	m	m

M = major, m = minor. With Spitzer, up to 50% of ES after auto-HCT is missed.

TA-TMA is diagnosed by biopsy or by ≥ 4 of 7 criteria at two time points

Criterion	Definition (harmonized criteria, Schoettler 2023)
Anemia	Hb fall of 1 g/dL, or new or persistent RBC transfusion need
Thrombocytopenia	No platelet engraftment, refractoriness, or 50% fall
Elevated LDH	Above the upper limit of normal for age
Schistocytes	Present
Hypertension	$\geq 140/90$ mmHg in adults; >99 th percentile if <18 years
Elevated sC5b-9	Above the upper limit of normal
Proteinuria	Random urine protein/creatinine ratio ≥ 1 mg/mg

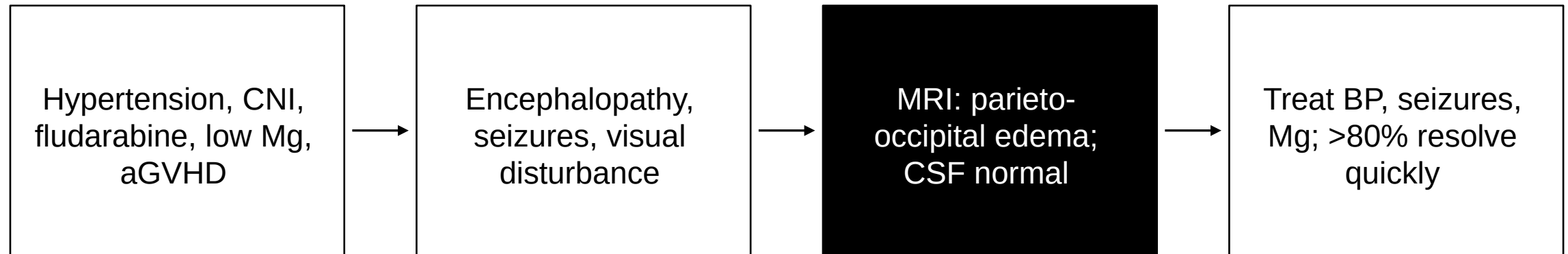
Clinical diagnosis: ≥ 4 of 7 within 14 days, at two consecutive time points.

Any high-risk feature makes TA-TMA a candidate for complement blockade

Feature	Standard risk	High risk (any one)
Peak LDH	<2× ULN	≥2× ULN
rUPCR	<1 mg/mg	≥1 mg/mg
Organ injury	KDIGO stage I AKI only	Any other organ dysfunction
Soluble C5b-9	Normal	Above ULN
Concurrent aGVHD or infection	—	aGVHD II–IV, or systemic infection
Management	Monitor closely; treat if cytopenia persists	Complement blockade (eculizumab, narsoplimab)

Survival: eculizumab ~70% (Jodele 2020); narsoplimab 68% at day 100 (Khaled 2022).

PRES is a reversible encephalopathy with posterior vasogenic edema



About 8% after allo-HCT (1% after haplo-HCT, 25% in sickle cell disease). Stopping the CNI did not change the course in the few comparative studies.

Question 1: SOS/VOD after allo-HCT

A 48-year-old woman with relapsed ALL received inotuzumab ozogamicin, then busulfan-based myeloablative allo-HCT. On day +14 she has painful hepatomegaly, ascites, 7% weight gain and bilirubin 4.2 mg/dL; symptoms began 3 days ago. Creatinine is 1.6× her baseline. What is the most appropriate management?

- A. Confirm HVP ≥ 10 mmHg before any specific therapy
- B. Supportive care; defibrotide only if bilirubin reaches 8 mg/dL
- C. Ursodeoxycholic acid alone until day +90
- D. Defibrotide for a fixed 5-day course
- E. Defibrotide 25 mg/kg/day now, for ≥ 14 –21 days until resolution

Answer E: start defibrotide now

E. Defibrotide 25 mg/kg/day now, for ≥ 14 –21 days until resolution

Bilirubin ≥ 2 mg/dL plus three findings = clinical SOS/VOD. Symptoms ≤ 4 days, weight gain $\geq 5\%$ and creatinine 1.5 – $2\times$ baseline put her in the severe category. Defibrotide is started as soon as the diagnosis is made, whatever the diagnostic category; HVPG is not required. UDCA is prophylaxis, not treatment.

Question 2: SOS/VOD in an infant

A 10-month-old boy with HLH receives busulfan-based MAC. Pre-HCT ultrasound showed hepatomegaly and minimal ascites. On day +9 he needs ≥ 1 weight-adjusted platelet transfusion daily without bleeding and has gained weight on 3 consecutive days despite furosemide. Bilirubin is 1.1 mg/dL. Which statement is correct?

- A. SOS/VOD cannot be diagnosed while bilirubin is < 2 mg/dL
- B. Right upper quadrant pain is required for the diagnosis
- C. He meets the pediatric EBMT criteria for SOS/VOD
- D. Preexisting hepatomegaly excludes SOS/VOD
- E. SOS/VOD can only be diagnosed after day +21

Answer C: pediatric EBMT criteria are met

C. He meets the pediatric EBMT criteria for SOS/VOD

pEBMT needs any 2 of 5 criteria, with no bilirubin threshold and no time limit. Refractory thrombocytopenia (≥ 1 weight-adjusted platelet transfusion/day) plus weight gain on 3 days despite diuretics fulfil it. A applies Baltimore/adult EBMT logic and misses anicteric SOS/VOD (~30% in children). Preexisting hepatomegaly only sets the baseline.

Question 3: Microangiopathy after allo-HCT

A 35-year-old man is on day +45 after matched unrelated donor allo-HCT, on tacrolimus plus sirolimus, with grade III gut aGVHD. Hb fell 2 g/dL, platelets are falling, LDH is 2.5× ULN, schistocytes are present, BP is 150/95 mmHg, rUPCR is 1.4 mg/mg and sC5b-9 is elevated. What is the best next step?

- A. Therapeutic plasma exchange as first-line therapy
- B. Observation, because this is standard-risk TA-TMA
- C. Complement blockade (e.g. eculizumab) and treatment of triggers
- D. Stop all immunosuppression immediately
- E. Withhold all platelet and red cell transfusions

Answer C: complement blockade for high-risk TA-TMA

C. Complement blockade (e.g. eculizumab) and treatment of triggers

He meets ≥ 4 of 7 harmonized criteria. LDH $\geq 2 \times$ ULN, rUPCR ≥ 1 mg/mg, elevated sC5b-9 and aGVHD II–IV are each high-risk features, so complement blockade is indicated, with aggressive treatment of GVHD. TPE is an alternative only when complement blockers are unavailable. Changes to immunosuppression must not provoke aGVHD.

References, part 1 of 2

1. Carreras E, Ruutu T, Mohty M, Corbacioglu S, de la Cámara R. Hepatic complications. In: Sureda A, Corbacioglu S, Greco R, Kröger N, Carreras E, editors. The EBMT Handbook: Hematopoietic Cell Transplantation and Cellular Therapies. 8th ed. Cham: Springer; 2024. p. 441–58. https://doi.org/10.1007/978-3-031-44080-9_49
2. Carreras E, Diaz-Ricart M, Jodele S, Penack O, Vasu S. Early complications of endothelial origin. In: Sureda A, Corbacioglu S, Greco R, Kröger N, Carreras E, editors. The EBMT Handbook: Hematopoietic Cell Transplantation and Cellular Therapies. 8th ed. Cham: Springer; 2024. p. 373–83. https://doi.org/10.1007/978-3-031-44080-9_42
3. Mohty M, Malard F, Alaskar AS, et al. Diagnosis and severity criteria for sinusoidal obstruction syndrome/veno-occlusive disease in adult patients: a refined classification from the European society for blood and marrow transplantation (EBMT). Bone Marrow Transplant. 2023;58:749. <https://doi.org/10.1038/s41409-023-01992-8>.
4. Spitzer TR. Engraftment syndrome following hematopoietic stem cell transplantation. Bone Marrow Transplant. 2001;27:893–8.
5. Maiolino A, Biasoli I, Lima J, et al. Engraftment syndrome following autologous hematopoietic stem cell transplantation: definition of diagnostic criteria. Bone Marrow Transplant. 2003;31:393–7.
6. Szmit Z, Gorczynska E, Król A, et al. Introduction of new pediatric EBMT criteria for VOD diagnosis: is it time-saving or money-wasting? Prospective evaluation of pediatric EBMT criteria for VOD. Bone Marrow Transplant. 2020;55:2138–46.
7. Yoon JH, Yoo KH, Sung KW, et al. Validation of treatment outcomes according to revised severity criteria from European Society for Blood and Marrow Transplantation (EBMT) for sinusoidal obstruction syndrome/veno-occlusive disease (SOS/VOD). Bone Marrow Transplant. 2019;54:1361–8.
8. Ruutu T, Juvonen E, Remberger M, et al. Improved survival with ursodeoxycholic acid prophylaxis in allogeneic stem cell transplantation: long-term follow-up of a randomized study. Biol Blood Marrow Transplant. 2014;20:135–8.

References, part 2 of 2

9. Corbacioglu S, Cesaro S, Faraci M, et al. Defibrotide for prophylaxis of hepatic veno-occlusive disease in paediatric haemopoietic stem-cell transplantation: an open-label, phase 3, randomised controlled trial. *Lancet*. 2012a;379:1301–9.
10. Grupp SA, Corbacioglu S, Kang HJ, et al. A phase 3, randomized, adaptive study of defibrotide (DF) vs best supportive care (BSC) for the prevention of hepatic veno-occlusive disease/sinusoidal obstruction syndrome (VOD/SOS) in patients (pts) undergoing hematopoietic cell transplantation (HCT): preliminary results. *Blood*. 2021;138:749.
11. Grant A, Chapman LRM, Mitchell R, O'Brien TA. Engraftment syndrome following hematopoietic stem cell transplant: a review of the literature. *Clin Transpl*. 2020;34:e13875.

12. Schoettler ML, Carreras E, Cho B, et al. Harmonizing definitions for diagnostic criteria and prognostic assessment of transplantation-associated thrombotic Microangiopathy: a report on behalf of the European Society for Blood and Marrow Transplantation, American Society for Transplantation and Cellular Therapy, Asia-Pacific Blood and Marrow Transplantation Group, and Center for International Blood and Marrow Transplant Research. *Transplant Cell Ther*. 2023;29:151–63.
13. Jodele S, Dandoy CE, Lane A, et al. Complement blockade for TA-TMA: lessons learned from a large pediatric cohort treated with eculizumab. *Blood*. 2020;135:1049–57.
14. Khaled SK, Claes K, Goh YT, et al. Narsoplimab, a Mannan-binding lectin-associated serine Protease-2 inhibitor, for the treatment of adult hematopoietic stem-cell transplantation-associated thrombotic Microangiopathy. *J Clin Oncol*. 2022;40:2447–57.

Children differ from adults in incidence, presentation and criteria

Feature	Children	Adults
Incidence	About 15–20%; up to 60% if high risk	About 5–10%
Late-onset (after day 21)	About 20%	Rare
Anicteric SOS/VOD	About 30%	About 15%
Bilirubin	Often preexistent or late; ≥ 2 mg/dL means severe disease	≥ 2 mg/dL mandatory for diagnosis
Hepatomegaly and ascites before HCT	Common; imaging baseline needed	—
Diagnostic criteria	Pediatric EBMT; refractory thrombocytopenia	Adult EBMT, Baltimore, Seattle

Ch. 49 gives adult incidence as 5–15% in the text and 5–10% in Table 49.4.

HBV needs antiviral prophylaxis; HCV and HAV are treated or deferred

Situation	Recommendation in the chapter
HBsAg+ or HBV-DNA+ recipient	Entecavir or tenofovir from before conditioning, ≥ 1 year
After stopping prophylaxis	Monitor HBV every 3 months; lamivudine not recommended
Anti-HBc+ only	ECIL: prophylaxis for all. EASL: only if risk $>10\%$
Pre-emptive strategy	HBsAg and HBV-DNA every 1–3 months; treat when detectable
HBsAg+ donor	Transmits in 22–44%; treat donor, prophylaxis for recipient
HCV-RNA+ recipient	Direct-acting antivirals (cure $>90\%$), ideally before HCT
HAV viremia, donor or recipient	Do not transplant: increased SOS/VOD risk

Beyond complement blockade, TA-TMA options rest on small series

Approach	Data in the chapter
Supportive care	Transfusions, stop culprit drugs, treat GVHD/infection, BP
Plasma exchange	Early TPE; option when complement blockers are unavailable
Rituximab + TPE	12/15 responses; useful with anti-factor H antibodies
Defibrotide	Responses in a retrospective multicenter review
In clinical trials	Ravulizumab, nomacopan (C5); pegcetacoplan (C3)
Prophylaxis	Defibrotide (high-risk children), N-acetylcysteine, statins

Hepatic GVHD and autoimmune hepatitis look alike; GGT and antibodies help

Feature	Autoimmune hepatitis	Hepatic GVHD
Other organs	Often asymptomatic; fatigue	Almost always GVHD elsewhere
AST rise	Moderate to severe	Minimal
GGT	Marked rise	Normal or decreased
Autoantibodies	AIH type 2 (anti-LKM, anti-LC1)	Usually type 1 pattern (ANA, ANCA)
Pathology	Portal infiltrate penetrating lobules	Bile duct damage, ductal degeneration
Steroid response	Excellent	Variable by severity